

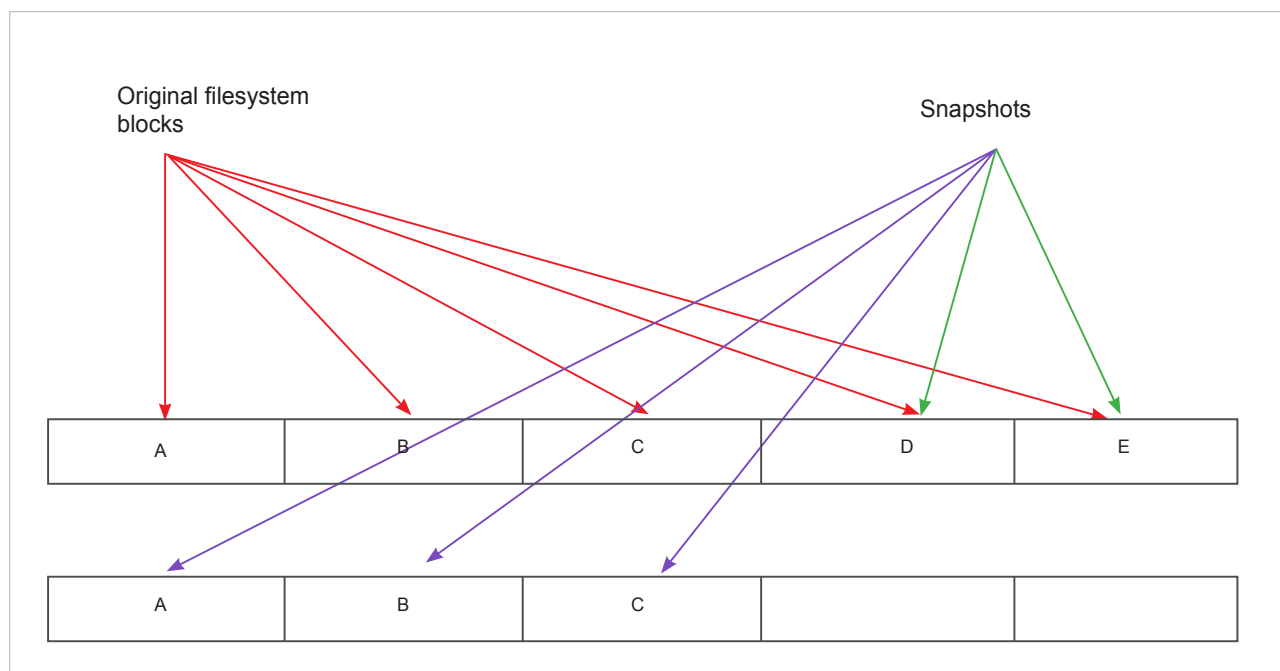


Figure 3: Modified blocks are first copied to the snapshot before being modified

Further, as shown in Figure 3, when the files on original file systems are to be modified, a copy-on-write mechanism ensures that the data blocks of files being edited are first copied to the snapshot data blocks before modifying the original data blocks. It is here that the data is copied block by block from the file system to snapshots. Over a period of time, when all the files on the original file systems have been modified, all its data blocks are copied to the snapshot before being overwritten. Hence, the snapshot consists of all the blocks of that data which existed on the file system at the time the snapshot was triggered. In other words, a snapshot is the image of the file system at that point of time when the snapshot was triggered.

Given that snapshots provide an alternate handle to access the file systems, the read requests from backup applications do not hinder the operations of production applications. Apart from backups, some of the additional benefits of a snapshot are:

- Ability to restore the file system to a previous state
- Reduction of load on original file system
- Test and development

Snapshots are also an important component because they fire the replications which play an important role in overall data protection implementation. Given the network bandwidth limitations, copying all the data over the network from the original file system is usually not feasible. A snapshot of the original file system is created, and the replication tool copies this data to the

replication site or on the cloud over the network. Once the replication is complete, subsequent data is then synced between the original and its replica. Features like compression and deduplication play a major role here to mitigate the network limitations.

While snapshots are a great data protection tool, they are not free from risks and limitations. Some of these are:

- As the snapshots refer to original file system data blocks, any corruption in the original files also corrupts the snapshot.
- Snapshots can fail to get restored.
- Multiple snapshots tax the original file system.
- The original file system is frozen for active IO when snapshots are taken. This delay blocks applications from accessing data. This might risk applications with intensive IOs.

OpenZFS used by many open source storage products, viz, FreeNAS/TrueNAS, has a well implemented snapshot mechanism. As it supports different commercial and open source backup software, FreeNAS is a fantastic choice for primary and secondary storage. **END** 🐧

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